Transformation migration to canonical core `5fe2c88`

STRATEGY-ID: HYMEKO-PLATFORM-2026-09-01-v1
ROADMAP-TASK: TRANSFORMATION-M2B

Created-at: 2026-09-02 18:40 CEDT

ETA

≈3–4 h: read-only audit (done) + frozen baseline (0.75) + mechanical migration of the write/id sites (1.0) + V&V, fault mutants, benchmark A/B (1.25) + report (0.5). Basis: the migration is *mechanical* (private-field writes → public constructors; `.0`→ `.raw()`); the cost is the byte-identical baseline + mutant net, not new logic.

1 Goal & scope

Re-pin `hymeko/transformation` (`hymeko_query`, `hymeko_formats`, `hymeko_emitter`) from `hymeko_core@15223bcd` to the canonical `hymeko_core@5fe2c88`, so the whole Cargo graph carries **exactly one** `hymeko_core`. No transform *semantics* change; query/transform/emit output stays **byte-identical** where the contract requires it. No `hymeko_core` production edit. No second IR / DTO / bridge created. Branch `feat/transformation-current-canonical-core`.

2 Why it was blocked, and why it is now feasible

core@main replaced the three public `decl_to_{node,edge,arc}: Vec<Option<Id>>` maps with a *private* 8-byte dispatch slot (`decl_to_record`, "option D", 2026-08-31). Transformation writes those maps directly at ≈**24 sites**. core@5fe2c88 exposes the matching **public construction seam**: `push_decl(DeclNode)->DeclId` (arena + NONE slot) and public `bind_{node,edge,arc}_decl(d, id)->Result<(),DispatchError>`. Every write site is *fresh construction* (a local `push_decl` helper + record binds), not in-place topology mutation, so it maps **1:1** to the public API. Audit bottom line: the migration needs **no missing core mutation API** and **no HARD STOP**.

3 Affected files (transformation only)

- `Cargo.toml` (workspace): `hymeko_core` rev `15223bcd`→`5fe2c88` (APPROVED-DEPENDENCY-EDIT: transformation-current-core-pin); parser unchanged.
- `hymeko_query/src/entropy.rs`, `rewrite/regen.rs`, `rewrite/split.rs`: `decl_to_*.push(None)` → folded into `ir.push_decl(..)`; `decl_to_X[did.raw()] = Some(id)` → `ir.bind_X_decl(did, id)?`.
- `hymeko_emitter/src/bridge.rs`: same, + `did.0` → `did.raw()` (≈14 `.0` sites across the tree).

- Any read-only accessor whose signature moved between `15223bcd` and `5fe2c88` (from the inventory) → the public getter.
- `Cargo.lock`: re-resolved; itemized diff audited (single `hymeko_core`).

No `hymeko_core/**` edit. If a minimal core seam turns out missing: document the smallest seam and HARD STOP (do not auto-edit core).

4 Test strategy

Phase-2 frozen baseline at `15223bcd`: full test/doctest counts; query / transform / emitted `.hymeko` + URDF/SDF/MJCF/DOT/Mermaid goldens; canonical hash; entity cardinalities; error variants; serde round-trip; the `compile/validate/transform/emit/query_batch` contract the Python binding consumes. Fixtures: simple hierarchy, import/namespace, annotations/values, $+/-/\sim$, duplicate incidence, weighted hyperarc, WebXR, VLAN/service, Steiner/sunflower/Mycielski, malformed source & malformed Ir; large workloads deterministically generated. Phase-4: re-run all against `5fe2c88`; assert byte-identical where the contract requires; fault mutants (old-Index hash, cross-version Ir, dropped namespace prefix, lost sign, deduped duplicate, partial transform commit, emitter reading another IR) each caught by a named test.

5 Performance budget

Peak RSS ≤ 16 GiB. Benchmark A/B (15223bcd vs 5fe2c88): compile, validate, query, query_batch, transform, emit, peak RSS. A $> 10\%$ regression blocks completion until profiled.

6 Rollback

Additive branch; revert = drop the branch. No core change to unwind; no main merge.

7 Risk anticipation

- **Worst case:** a deterministically-generated $\geq 10^5$-incidence source through `compile`→`transform`→`emit`; the migrated construction path must stay $O(\text{nnz})$ and byte-identical.
- **Contract:** `bind_*_decl` returns `Result`; the old field write could not fail. The migration must handle/propagate `DispatchError` (index-overflow / double-bind) rather than `unwrap` silently at the 10^5 scale — but a bind that the old code did unconditionally is provably first-and-only, so a documented invariant + `expect` is acceptable where proven, else propagate.
- **Two core versions:** `cargo tree` must show exactly one `hymeko_core`; any second (via a stale transitive pin) is a build-breaking finding.
- **Dirty user checkout:** the user is actively on `feat/geometry-...` with new `hymeko_dynamics`/`hymeko_opti` crates. Work only in the clone `/d/hnnwork/transformation`; never touch the user checkout.

8 Dependency repair continuation (2026-09-10)

Created-at: 2026-09-10T01:50:53.773910+02:00. ETA 10–20 minutes: manifest alignment 3, isolated compilation/tests 10, publication/archive 5. User requests fixing and publishing the active Transformation checkout. Backport the already published main dependency configuration to the active authoring branch e76562f; main 6e70c28 is three commits ahead, while dirty source work prevents an automatic whole-tree fast-forward. No stash, branch switch or new worktree. Preserve other staged/unstaged changes; back up the locally overridden Cargo.lock before reconciling it. Affected files: workspace Cargo.toml/lock, Dynamics and Geometry Cargo.toml, the four legacy Query reference fixtures in entropy.rs, rewrite/regen.rs and rewrite/split.rs, this plan bundle. The full workspace build also exposes legacy RefAtomR constructors in Emitter src/bridge.rs, tests/test_bridge.rs and WebXR tests/mod.rs; use Core plain constructors and assert unspecified direction for all three signs in the existing bridge round-trip test. These additional compatibility fixes must also be merged into main after branch validation. Optimize tests/authority.rs replaces its blanket patch-section rejection with a check that all Core declarations use the same published exact Git identity; the existing dirty formatting in this file is excluded from staging. Reuse Core e1ebb415, Graph 8f44b8ca, Parser 0019eb0a and Tensor 14c1d231, the tested main set. No Core production edit or new API. Intra-repository workspace paths remain; external siblings become exact Git dependencies. Validate the staged publication through an index export below ignored target, running Cargo outside the parent configuration, shared-owner tests and dependency identity audit. Do not publish unrelated local source edits. Cap each test process at 180 seconds and four build jobs; no new performance campaign for manifest-only alignment. Existing main integration evidence supplies runtime baseline. Publish the active authoring branch to both remotes; main already contains these dependency changes. Rollback only task hunks or the focused commit, preserving the pre-task lock backup. Report exact branch, SHA and any pre-existing test limitations; archive reports through the shared provider.